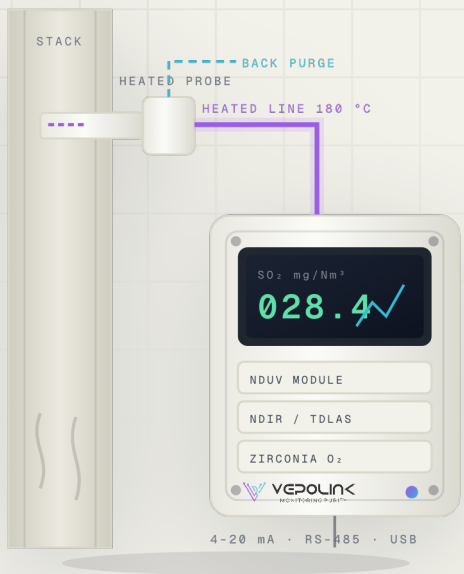
VEPOLINK
MONITORING PURITY

ENVIRONMENTAL INTELLIGENCE × EMISSION MONITORING

CONTINUOUS EMISSION MONITORING · CEMS

Extractive Flue / Stack Gas Analyzer *The stack, reported honestly*

A heated extractive CEMS built on NDUV, NDIR, FTIR, TDLAS, DOAS and zirconia measurement — SO_2, NOx, CO, CO_2, O_2, NH_3, HCl, HF and Cl_2 measured continuously, down to 0.1 ppm, in harsh industrial service.



0.1 ppm
DETECTION LIMIT, SO_2 & NOX

2 %
ACCURACY · LINEARITY · DRIFT

200 °C
HEATED SAMPLING PATH

IP65 ENCLOSURE | **AUTO** ZERO & SPAN | **RS-232 / RS-485** · 4-20 MA

2%	2%	2%	2%
ACCURACY	LINEARITY	SPAN DRIFT	ZERO DRIFT

01 OVERVIEW

Continuous stack *emission monitoring.*

Flue gas is hot, wet, acidic and full of dust — and the number you report has to survive all of it. An extractive CEMS solves that by keeping the sample above its dew point from the probe to the cell, filtering it in two stages, and measuring it in an instrument that lives in a clean, conditioned cabinet rather than on the stack.

The Vepolink Extractive Flue / Stack Gas Analyzer is designed for continuous monitoring of industrial emissions. It uses advanced UV, NDIR, FTIR, TDLAS and DOAS technologies to deliver accurate measurement of multiple gases while ensuring reliable operation in harsh industrial environments. NDUV with PLS spectral deconvolution takes SO_2 and NOx down to 0.1 ppm, NO and NO_2 are measured directly without a converter, and a pulsed xenon source with no moving optical components keeps the optics stable between services.

AT A GLANCE

- EXTRACTIVE
- NDUV + PLS
- NDIR · FTIR
- TDLAS · DOAS
- ZIRCONIA O_2
- HEATED TO 200 °C
- AUTO ZERO & SPAN
- IP65

· PRODUCT FEATURES

NDUV with PLS

Non-dispersive UV and partial least squares deconvolution give ultra-low detection limits — SO_2, NOx and other gases down to 0.1 ppm on a high-resolution diode array detector.

Pulsed Xenon, No Moving Optics

A long-life pulsed xenon lamp and a fixed optical bench — nothing rotates, nothing chops — so drift stays at 2 % and reliability stays high.

Direct NO and NO_2

Both species are measured directly, with no NO_2-to-NO converter to maintain or to bias the NOx total when its efficiency falls.

Modular and Self-Calibrating

Measurement modules pull out for service; automatic and remote zero and span calibration keep the system in specification without a site visit.

02 WORKING PRINCIPLE

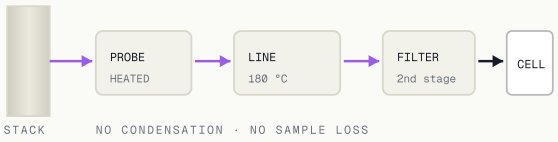
Extract hot, measure *clean.*

A heated sampling probe extracts flue gas through a heated sample line and filtration system before it enters the analyzer. The heated path prevents condensation, while the UV/IR analyzer continuously measures gas concentrations with high precision.

SAMPLING

Heated Path, Two-Stage Filter

PROBE · HEATED LINE · SECONDARY FILTRATION



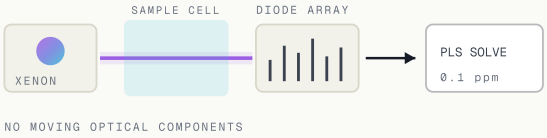
The probe filter takes the dust, the heated line holds the sample above its acid and water dew point, and a secondary filter protects the analyzer module. An automatic back purge clears the probe on schedule.

`Sample arrives as it left the stack`

NDUV + PLS

SO_2, NO, NO_2 and NH_3

XENON SOURCE · DIODE ARRAY · 0.1 PPM



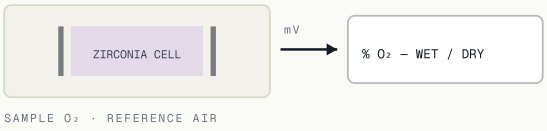
Pulsed UV light crosses the sample cell onto a high-resolution diode array. Partial least squares resolves overlapping absorption structure, so each species is reported separately at very low concentration.

`Full spectrum, not one band`

ZIRCONIA

Oxygen Reference

O_2 FOR CORRECTION TO REFERENCE CONDITIONS



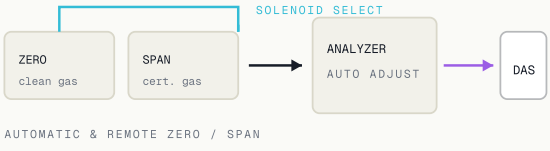
A zirconia cell compares sample oxygen against reference air and generates a millivolt signal. Oxygen is both a combustion indicator and the basis for correcting emissions to a reference O_2 level.

`Corrects every reported value`

CALIBRATION

Automatic and Remote

ZERO GAS · CERTIFIED SPAN GAS · PRESSURE COMP.



Solenoids introduce zero and span gas on a timed cycle and the analyzer adjusts itself, logging every check. Pressure compensation keeps readings accurate as stack and ambient conditions change.

`Verified without a site visit`

03 GASES & TECHNOLOGY

One cabinet, *the right technique per gas.*

Each component is measured by the method that suits it best, and the selection is configured for the plant. Measuring ranges are set to the emission limits and expected concentrations of the application and confirmed at order.

GAS	NAME	TECHNOLOGY
SO_2	Sulphur Dioxide	NDUV · NDIR · FTIR
NO	Nitric Oxide	NDUV · NDIR · FTIR
CO	Carbon Monoxide	NDUV · NDIR · FTIR
CO_2	Carbon Dioxide	NDIR · TDLAS · FTIR
O_2	Oxygen	Zirconia

GAS	NAME	TECHNOLOGY
NH_3	Ammonia	NDUV · TDLAS · FTIR
HCl	Hydrogen Chloride	TDLAS · IR DOAS
HF	Hydrogen Fluoride	TDLAS
Cl_2	Chlorine	DOAS IR
NOx	Nitrogen Oxides	NO + NO_2, direct

MONITORED PARAMETERS

GAS, PARTICULATE AND FLUE CONDITIONS ON ONE SYSTEM

SO_2	NO	NO_2	NOx	CO
CO_2	O_2	NH_3	HCl	HF
Cl_2	Dust	Humidity	Temperature	Flow

STEP 01

Extract

The heated probe draws flue gas from the stack through a coarse filter, with automatic back purge to clear dust.

STEP 02

Transport

A heated sample line holds the gas above dew point all the way to the shelter, so nothing condenses or is lost.

STEP 03

Condition

Secondary filtration and pressure compensation deliver a clean, stable sample to the measurement modules.

STEP 04

Report

Concentrations go to the data acquisition system over RS-232, RS-485, USB or 4–20 mA, corrected to reference O_2.

04 SPECIFICATIONS

Technical *specifications.*

MEASUREMENT

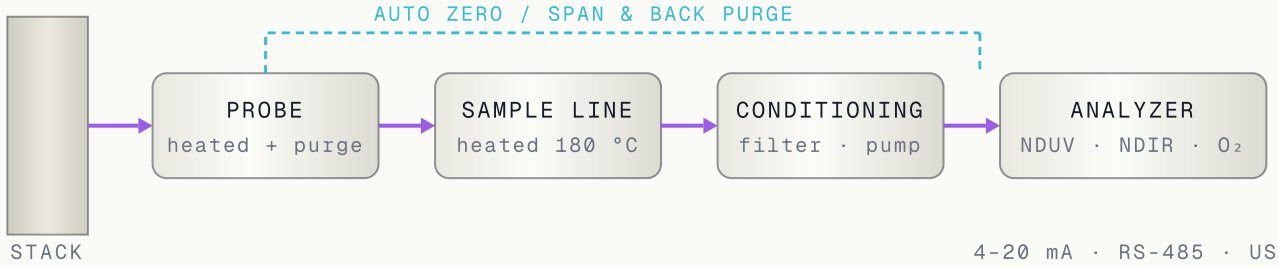
System Type	Extractive CEMS, heated
Technologies	NDUV, NDIR, FTIR, TDLAS, DOAS, zirconia O_2
Measured Gases	SO_2, NO, NO_2, NOx, CO, CO_2, O_2, NH_3, HCl, HF, Cl_2
Other Parameters	Dust, humidity, temperature, flow
Detection Limit	Down to 0.1 ppm, SO_2 & NOx
Accuracy	2 %
Linearity	2 %
Zero Drift	2 %
Span Drift	2 %
Detector	High-resolution diode array
Light Source	Pulsed xenon lamp, long life

SAMPLING · OUTPUT · BUILD

Sampling	Heated probe, heated line, secondary filtration
Sample Temperature	50 - 200 °C heated path
Probe Purge	Automatic back purge
Calibration	Automatic & remote zero / span
Compensation	Pressure compensation
Analog Output	4-20 mA
Digital Interfaces	RS-232 · RS-485 · USB
Integration	SCADA / DAS over serial and analog
Construction	Modular measurement bays
Ingress Protection	IP65
Operation	Continuous, online

SYSTEM ARCHITECTURE

STACK → PROBE → HEATED LINE → CONDITIONING → ANALYZER



05 INSTALLATION & UPKEEP

Installed as a system, *not a box.*

A CEMS is only as good as its sampling. Choose a **representative sampling plane** on a straight run of stack, keep the **entire path heated** from probe to cell, and let the **automatic zero and span cycle** prove the instrument daily. Everything the regulator sees depends on those three things.

01 · SAMPLING PLANE

Representative section

A straight run of stack with well-developed flow, clear of bends, dampers and dilution, following the applicable siting standard.

02 · PROBE

Heated, filtered, purged

Heated probe with a coarse filter at the flange and automatic back purge to shed dust without stopping monitoring.

03 · SAMPLE LINE

Heated to 200 °C

The whole transport path is held above the water and acid dew point so soluble gases such as SO_2, HCl and NH_3 arrive intact.

04 · SHELTER

Conditioned cabinet

The analyzer, pump, filters and calibration gases sit in an IP65 enclosure or shelter, away from stack heat and vibration.

05 · CALIBRATION

Automatic and remote

Timed zero and span checks with certified gas, adjustable and reviewable remotely, with every check recorded.

06 · SERVICE

Modular swap

Measurement modules, filters and the xenon lamp are replaceable items; the sampling system is the routine maintenance item.

EXTRACTIVE VERSUS IN-SITU

In-situ analyzers avoid the sample line, but the extractive approach puts the optics in a clean, temperature-controlled box where calibration gas can be applied through the whole path. For multi-component regulatory reporting — SO_2, NOx, CO, HCl, NH_3 and O_2 together — that traceable, self-verifying arrangement is usually what the authority expects. Where a single component in a hostile duct is the requirement, our cross duct TDLAS analyzer is the counterpart.

DATA AND REPORTING

Outputs integrate with SCADA and data acquisition systems over RS-232, RS-485 and 4–20 mA, with USB for local download. Concentrations can be reported corrected to reference oxygen and standard conditions, together with dust, flow, humidity and temperature, as a complete emission record.

06 APPLICATIONS

Where it *works.*

Wherever a stack is subject to a consent limit — and wherever the combustion behind it is worth optimising.

Coal Fired Power Plants	Waste-to-Energy Plants	Cement Plants
Glass Manufacturing	Lime Industries	Steel & Sinter Plants
Coke Oven Plants	Boiler Emission Monitoring	Industrial Process Monitoring

COMPLIANCE AND CONTROL TOGETHER

The same measurements that satisfy a consent limit also run the plant. Oxygen and CO set excess air on a boiler; ammonia slip governs SCR reagent dosing; HCl drives scrubber lime feed. A CEMS that is trusted by the operator, not just the inspector, pays back in fuel and reagent as well as in compliance.

SPECIFYING A SYSTEM

Tell us the components and limits you must report, the stack conditions — temperature, moisture, dust loading, pressure — the distance from the sampling plane to the shelter, and the reporting system you feed. We return the measurement technique per gas, the sampling and calibration arrangement, the shelter and the output configuration as one package.

07 FAQ

Frequently asked *questions.*

Q What is an extractive flue stack gas analyzer?

An online gas analyzer that continuously monitors industrial stack emissions using an extractive sampling system — the gas is drawn from the stack and measured in an analyzer cabinet.

Q Which gases can be monitored?

SO_2, NOx, CO, CO_2, O_2, NH_3, HCl, HF, Cl_2 and other gases depending on the application, alongside dust, humidity, temperature and flow.

Q Does it support remote calibration?

Yes. The analyzer supports automatic and remote zero and span calibration, with each check logged.

Q Can it integrate with SCADA?

Yes, through RS-232, RS-485 and analog 4–20 mA outputs, with USB available locally.

Q How low can it measure?

NDUV with PLS deconvolution takes SO_2, NOx and other gases down to 0.1 ppm, using a high-resolution diode array detector.

Q Is an NO_2 converter required?

No. NO and NO_2 are measured directly, so there is no converter to maintain and no efficiency loss biasing the NOx total.

Q Why is the sampling system heated?

To prevent condensation. Water forming in the probe or line would absorb SO_2, HCl and NH_3 and the reading would understate the true emission.

Q How is the probe kept clear?

An automatic back purge system blows the probe filter clear on a schedule, and secondary filtration protects the analyzer module downstream.

Q How stable is the measurement?

Accuracy, linearity, zero drift and span drift are each 2 %, helped by a pulsed xenon source and an optical bench with no moving parts.

Q What maintenance does it need?

Routine attention to the sampling system — probe filter, line and secondary filter — plus periodic calibration gas replacement. Measurement modules are replaced as units.

This brochure is general product information. Measurement technique per gas, measuring ranges, sampling arrangement and performance figures are indicative and confirmed at order against the stack conditions and the reporting requirements of the site. Siting, calibration and reporting must follow the applicable national standards and the consent conditions of the installation. Specifications may change without notice as the product is improved.

VEPOLINK
MONITORING PURITY

CONTINUOUS, CORRECTED, ON THE RECORD

Report the stack *with confidence.*

Send us the components and limits you report, the stack conditions and the distance to your shelter — we'll return the technique per gas, the sampling system and the outputs as one package. Pick how you'd like to start.

01 **Talk to our engineers**
Speak directly to the team that designs, installs and commissions CEMS. No call centre.
+91 99538 86178 →

02 **Request a CEMS configuration**
RECOMMENDED
Share your consent limits, stack conditions and layout — we'll spec gases, techniques, sampling, shelter and reporting.
sales@vepolink.com →

03 **Book a demonstration**
See a live analyzer, an automatic calibration cycle and data flowing to a DAS with our engineering team.
Request a demo →

WWW.VEPOLINK.COM · SALES@VEPOLINK.COM